

Individual-Assignment-5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪ Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))

```

Class Code

```

# ---- sites object ----

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# ---- cleaning taxon_list ----

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# ---- cleaning water_quality ----

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames

```

```

unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# ---- cleaning aq_ins ----

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)

```

```

summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site (ordered by median salinity, descending)
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↵ = TRUE),
  site_full = fct_rev(site_full))

# ---- salinity plot ----

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent medians
stat_summary(fun = median,
  color = "red",
  size = 1) +

```

```

# relabelling x- and y-axes
labs(x = "Site",
      y = "Median salinity (ppt)",
      title = "Site salinity") +
# different theme
theme_bw()

# ---- copepods plot ----

# creating new object from clean aquatic inverts
copepods_df <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods_df,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                  y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Copepods") +
# different theme
theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: log_ave_lit
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

b. Wrangle your data

```
# creating new object from clean aquatic inverts for ostracods
ostracods_df <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows to check output
slice_sample(ostracods_df, n = 10)
```

```
# A tibble: 10 x 24
```

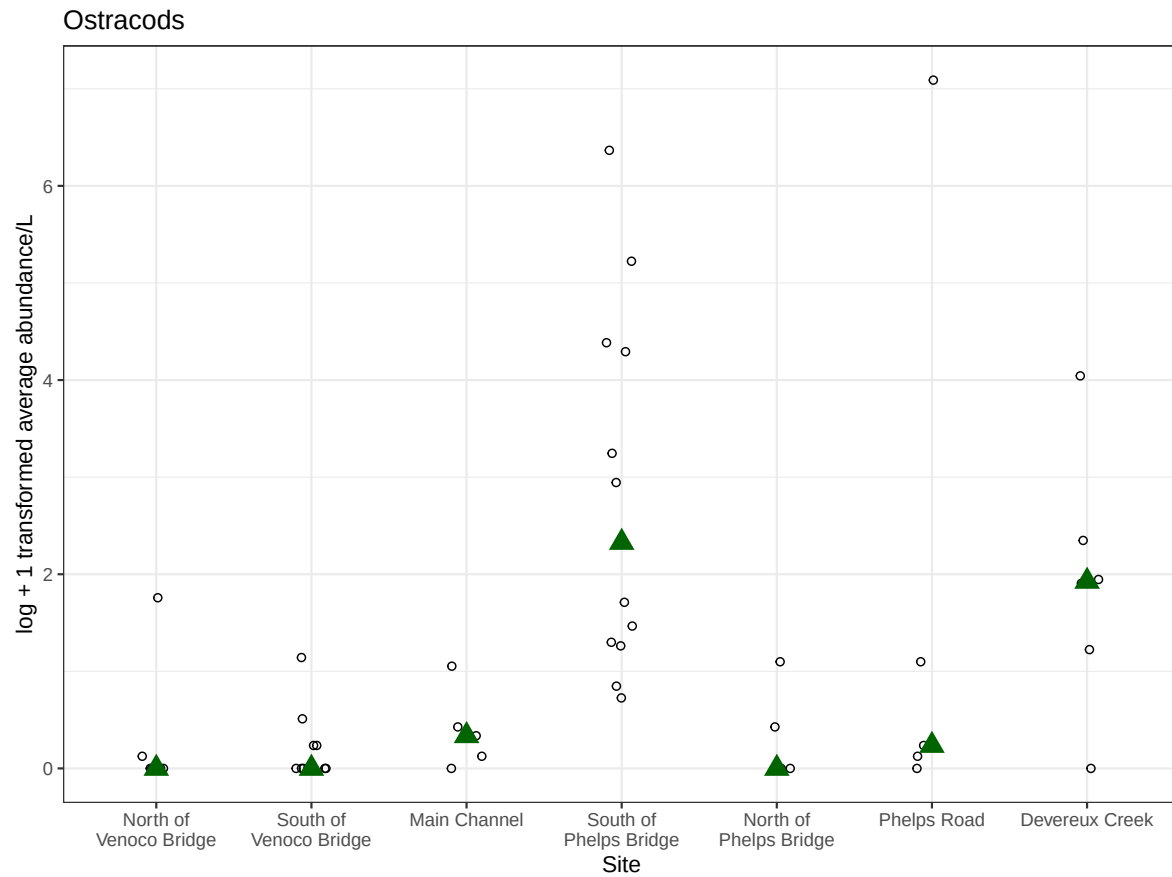
	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
2	2023-08-12_NEC	2023-08-12 00:00:00	NEC	ostr~	0	NA	NA	NA
3	2023-03-03_NDC	2023-03-03 00:00:00	NDC	ostr~	5.73	7.45	0.688	5.64
4	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA
5	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	ostr~	0	9.72	74.2	11.4
6	2022-08-12_NEC	2022-08-12 00:00:00	NEC	ostr~	0	7.37	NA	NA
7	2023-07-17_NPB	2023-07-17 00:00:00	NPB	ostr~	4.53	NA	NA	NA
8	2023-11-11_NEC	2023-11-11 00:00:00	NEC	ostr~	0	NA	38.6	0.25
9	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.4	6.53	0.3	7.6
10	2022-02-21_NPB	2022-02-21 00:00:00	NPB	ostr~	1.33	7.39	34.1	0.34

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

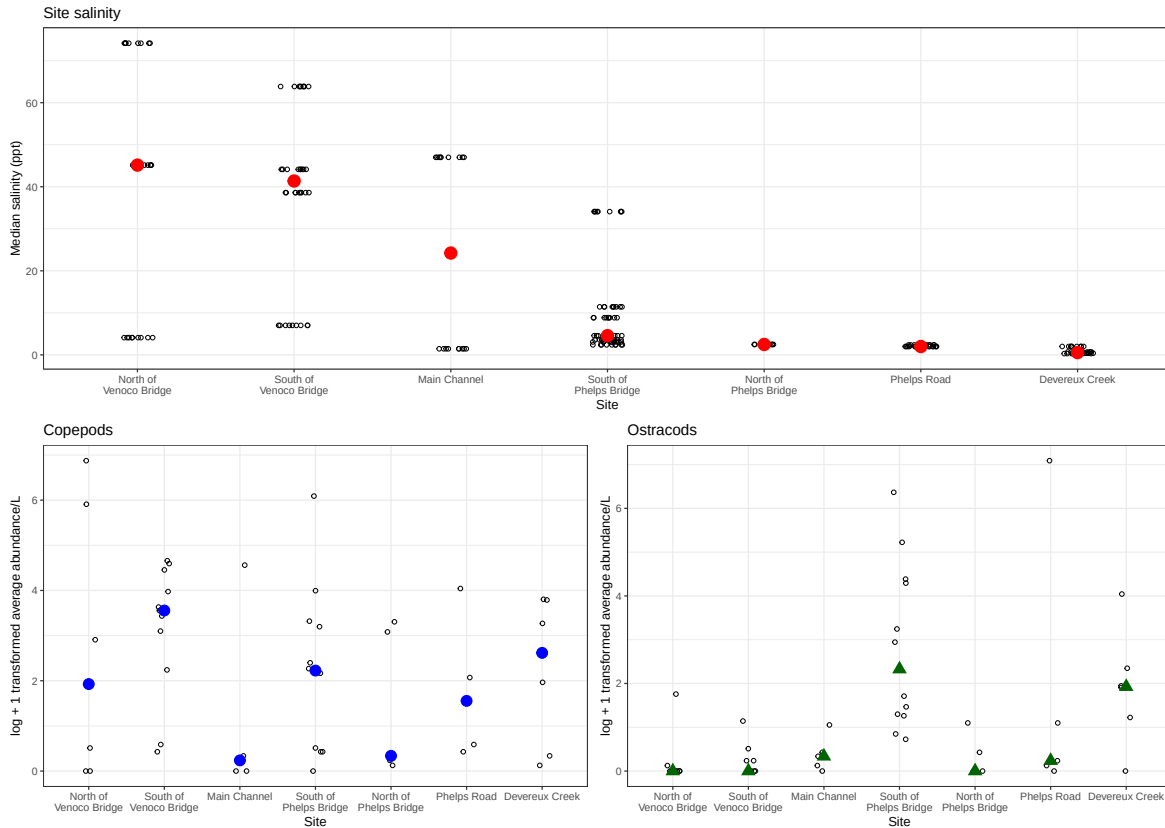
```
# base layer
ostracods <- ggplot(data = ostracods_df,
  # x-axis: site
  # y-axis: log + 1 transformed average abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: individual average abundance/L measurements
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# second layer: large filled orange triangle for median - distinct color,
# shape, and size from individual points
stat_summary(geom = "point",
  fun = "median",
  size = 4,
  color = "darkgreen",
  shape = 17) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# axis labels and title indicating taxonomic group
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Ostracods") +
# non-default theme
theme_bw()

# displaying object
ostracods
```



d. Create a multipanel figure

```
# combining salinity, copepods, and ostracods (bottom right) using patchwork
salinity / (copepods | ostracods)
```



e. Interpret your figure

Copepod median abundance is highest at South of Venoco Bridge with a little over 3 log-units. However the Main Channel and North of Phelps Bridge have medians near 0 log-units. Ostracod median abundance is highest at South of Phelps Bridge with a little over 2 log-units and second highest at Devereux Creek with 2 log-units. However, most other sites showing medians near 0 log-units.

Sites with the highest median salinity such as North and South of Venoco Bridge have ostracod medians near 0, suggesting ostracods are not best suited for high-salinity conditions. The highest ostracod medians occur at South of Phelps Bridge and Devereux Creek, which have lower median salinities (~5 and ~1 ppt), demonstrating a possible preference for water with lower salinity.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

```
# creating new object for abundance vs. salinity visualization
sal_abund_df <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon names to title case for readable facet labels
  mutate(taxon = str_to_title(str_replace_all(taxon, "_", " "))) |>
  # reorder taxon factor levels by maximum abundance (descending)
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = "max", .desc = TRUE)) |>
  # keep only columns needed for the plot
  select(taxon, med_sal, log_ave_lit, ave_lit)

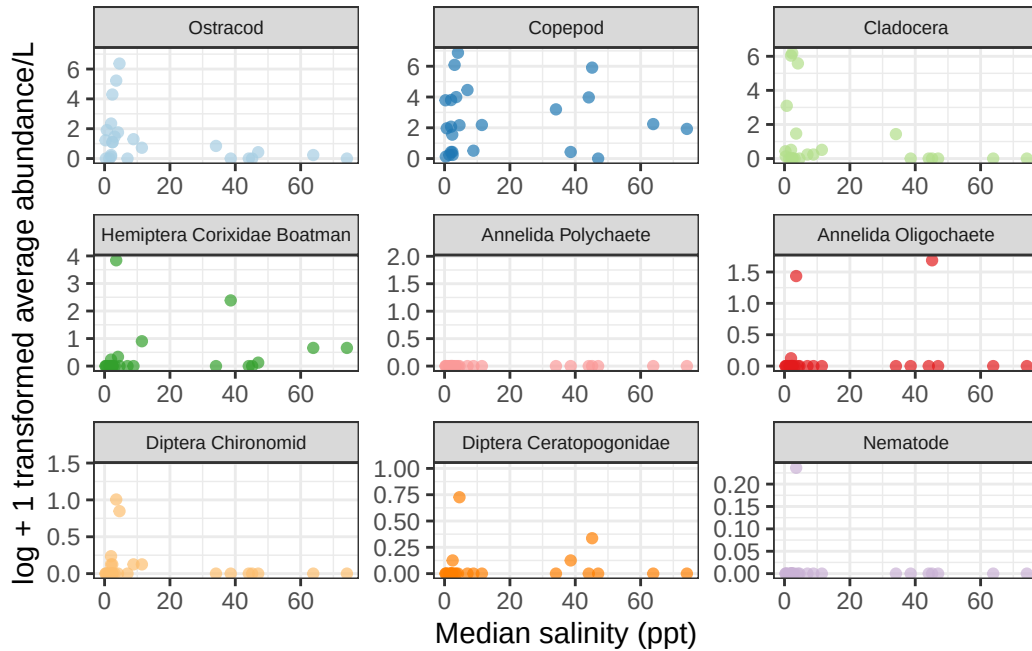
# display 10 random rows to check output
slice_sample(sal_abund_df, n = 10)
```

A tibble: 10 x 4

	taxon	med_sal	log_ave_lit	ave_lit
	<fct>	<dbl>	<dbl>	<dbl>
1	Annelida Polychaete	NA	0	0
2	Diptera Chironomid	1.98	0.236	0.267
3	Nematode	38.6	0	0
4	Cladocera	NA	0	0
5	Ostracod	NA	0	0
6	Diptera Chironomid	3.57	1.01	1.73
7	Nematode	NA	0	0
8	Copepod	NA	0.511	0.667
9	Diptera Chironomid	1.99	0.125	0.133
10	Hemiptera Corixidae Boatman	NA	0.125	0.133

c. Code your figure

```
# base layer
ggplot(data = sal_abund_df,
       # x-axis: median salinity
       # y-axis: log + 1 transformed average abundance
       # color: taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
# point layer for each observation
geom_point(alpha = 0.7,
           size = 1.5) +
# separate panel for each taxon with free scales on both axes
facet_wrap(~ taxon,
           scales = "free") +
# readable axis labels
labs(x = "Median salinity (ppt)",
     y = "log + 1 transformed average abundance/L") +
# non-default color palette
scale_color_brewer(palette = "Paired") +
# remove legend
theme_bw() +
theme(legend.position = "none",
      # improve readability of facet strip text
      strip.text = element_text(size = 7))
```



d. Interpret your figure

Copepods and ostracods show the broadest range of abundances and are present across the full salinity gradient (0–75 ppt), with their highest abundances concentrated at low salinities (below ~10 ppt) but with some observations persisting at 40–75 ppt, suggesting relatively high salinity tolerance compared to other taxa. Annelida Oligochaete is consistently near-zero in abundance across all salinities, with only a few low-abundance observations clustered at low salinities, implying it is rare in this system regardless of salinity. Annelida Polychaete and Nematodes have the smallest overall abundance, with almost all observations clustered near zero across all salinity levels, suggesting they are essentially absent from the samples at NCOS sites.